

“AI-Powered Humanoid Robot Fleet Management System”

Team: Robo-Nexus

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GitHub Repository:

<https://github.com/sfsu-joseo/csc675-775-database-api-project-robo-nexus>

Version	Description
V1	Initial Submission

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Normalization Table

Entity	Failed NF	Reason	Fixes
Robot	3NF	Manufacturer depends on robot model id not on robot id, and other ids were creating transitive dependency. FD1: RobotID->RobotModelID FD2: RobotModelID->ManufactureID RobotModelID is not a superkey. And ManufactureID is not a prime.	remove manufacturer_id, Performance_metric_id, Charging_station_id,Alert_id from robot table.
DeploymentLocation	3NF	Technician_id implies a single one for any deployment location, charging session_id creates transitive dependency because chargingSession already references DeploymentLocation. FD1: Technician_id->	Remove technician_id,Charging_Session_id

		Deployment_location_id FD2: Charging_Station_id -> Deployment_location_id For both cases, deployment_location_id is the superkey meaning Y->X instead of X->Y.	
Technician	1NF	Name is not an atomic attribute.	Create a different table with names.
SupportRequest	3NF	We had FD1: SupportRequestID-> RobotID, UserID and also there is functional dependency in RobotID-> UserID. In this table, RobotID is not a superkey and UserID is not a prime attribute.	Removed User_ID
User_Deployment_Location	1NF	Missing Primary Key	Added UserDeploymentLocationID
Emotional_Analyze	1NF	Missing Primary Key	Added EmotionalAnalyzeID

PerformanceDiagnostic	1NF	Missing Primary Key	Added PerformanceDiagnosticID
EmotionalInteractionSession	1NF	Missing Primary Key	Added EmotionalInteractionSessionID
Technician_Support_Request	1NF	Missing Primary Key	Added TechnicianSupportRequestID
RobotChargingStation	1NF	Missing Primary Key	Added RobotChargingStationID
logSession	1NF	Missing Primary Key	Added logSessionID
HardwareMaintenance	1NF	Missing Primary Key	Added HardwareMaintenanceID
ChargingStationSession	1NF	Missing Primary Key	Added ChargingStationSessionID
PatientEmotionalRecord	1NF	Missing Primary Key	Added PatientEmotionalRecordID

List of Main Data Items and Entities (Normalize)

1. **Robot:** Strong
 - 1.1 robot_id: key, numeric (PK)
 - 1.2 ~~charging_station_id (FK)~~ (removed)
 - 1.3 robotmodel_id: numeric (FK)
 - 1.4 deployment_location_id: numeric (FK)
 - 1.5 ~~alert_id: numeric (FK)~~ (removed)
 - 1.6 user_id: numeric (FK, optional)
 - 1.7 software_version_id: numeric (FK)
 - 1.8 ~~manufacturer_id: numeric (FK)~~ (removed)
 - 1.9 ~~performance_metric_id: numeric (FK)~~ (removed)
 - 1.10 robotrole_id: numeric (FK)
2. **RobotModel:** Strong
 - 2.1 robotmodel_id: key, numeric (PK)
 - 2.2 manufacturer_id: numeric (FK)
 - 2.3 model: enum, alphanumeric
3. **User:** Strong
 - 3.1. user_id: key, numeric (PK)
 - 3.2. user_name: alphanumeric
4. **Manufacturer:** Strong
 - 4.1 manufacturer_id: key, numeric (PK)
 - 4.2 name: enum, alphanumeric
5. **Hardware:** Strong
 - 5.1 hardware_id: key, numeric (PK)
 - 5.2 manufacturer_id: numeric (FK)
6. **RobotHardware:** Weak
 - 6.1 robot_hardware_id: key, numeric (PK)
 - 6.2 robot_id: numeric (FK)

- 7. **SoftwareVersion:** Strong
 - 7.1 software_version_id: key, numeric
- 8. **DeploymentLocation:** Strong (Superclass)
 - 8.1 deployment_location_id: key, numeric (PK)
 - 8.2 ~~technician_id: numeric (FK)~~
 - 8.3 address_id: numeric (FK)
 - 8.4 environment_type: enum, alphanumeric
(warehouse/hospital/home/factory)
- 9. **Address:** Strong
 - 9.1 address_id: key, numeric (PK)
 - 9.2 street: alphanumeric
 - 9.3 city: alphanumeric
 - 9.4 state: alphanumeric
 - 9.5 zipcode: alphanumeric
- 10. **Task:** Strong
 - 10.1 task_id: key, numeric (PK)
 - ~~10.1 ai_response_id: numeric (FK)~~
 - 10.2 task_name: alphanumeric
 - 10.3 description: text
 - 10.4 priority: enum, numeric
 - 10.5 status: bool (Complete)
- 11. **TaskAssignment:** Weak
 - 11.1 task_assignment_id: key, numeric (PK)
 - 11.2 robot_id: numeric (FK)
 - 11.3 user_id: numeric (FK)
 - 11.4 task_id: numeric (FK)
 - 11.5 assigned_at: timestamp
- 12. **TaskExecution:** Weak
 - 12.1 task_execution_id: key, numeric (PK)
 - 12.3 task_id: numeric (FK)
 - 12.4 start_time: timestamp

- 12.5 end_time: timestamp
- 12.6 status: bool, (Successful)
- 13. **EmotionalRecord**: Weak
 - 13.1 emotional_id: key, numeric (PK)
 - 13.2 pattern_id: numeric (FK)
 - 13.3 emotional_state: enum, alphanumeric
 - 13.4 timestamp: timestamp
 - 13.5 description: text
- 14. **InteractionSession**: Weak
 - 14.1 interaction_id: key, numeric (PK)
 - 14.2 robot_id: numeric (FK)
 - 14.3 user_id: numeric (FK)
 - 14.4 home_id: numeric (FK)
 - 14.4 start_time: timestamp
 - 14.5 end_time: timestamp
- 15. **RobotRole**: Strong
 - 15.1 robotrole_id: key, numeric (PK)
 - 15.2 user_id: numeric(FK)
 - 15.3 robotrole: enum, alphanumeric
- 16. **EnergyStatus**: Weak
 - 16.1 energy_status_id: key, numeric (PK)
 - 16.2 robot_id: numeric (FK)
 - 16.3 charging_session_id: numeric (FK)
 - 16.4 battery_level: numeric
 - 16.5 timestamp: timestamp
- 17. **PerformanceMetric**: Weak
 - 17.1 performance_metric_id: key, numeric (PK)
 - 17.2 robot_id: numeric (FK)
 - 17.3 metric_type: alphanumeric
 - 17.4 metric_value: numeric
 - 17.5 timestamp: timestamp

- 18. **Alert:** Weak
 - 18.1 alert_id: key, numeric (PK)
 - 18.2 robot_id: numeric (FK)
 - 18.3 severity: enum, alphanumeric
 - 18.4 timestamp: timestamp
- 19. **DiagnosticReport:** Weak
 - 19.1 diagnostic_report_id: key, numeric (PK)
 - 19.2 robot_id: numeric (FK)
 - 19.3 ai_model_id: numeric (FK)
 - 19.4 generated_at: timestamp
 - 19.5 summary: text
- 20. **Log:** Weak
 - 20.1 log_id: key, numeric (PK)
 - 20.2 robot_id: numeric (FK)
 - 20.3 ~~deployment_location_id: numeric (FK, optional)~~
 - 20.4 CreatedAt: timestamp
 - 20.5 event_type: alphanumeric
 - 20.6 event_details: text
- 21. **Sensor:** Weak
 - 21.1 sensor_id: key, numeric (PK)
 - 21.2 robot_id: numeric (FK)
 - 21.3 sensor_type: alphanumeric
 - 21.4 status: enum, alphanumeric
- 22. **ChargingStation:** Strong
 - 22.1 charging_station_id: key, numeric (PK)
 - 22.2 station_name: alphanumeric
 - 22.3 capacity: numeric
 - 22.4 status: enum, alphanumeric
- 23. **ChargingSession:** Weak
 - 23.1 charging_session_id: key, numeric (PK)
 - 23.2 deployment_location_id: numeric (FK)

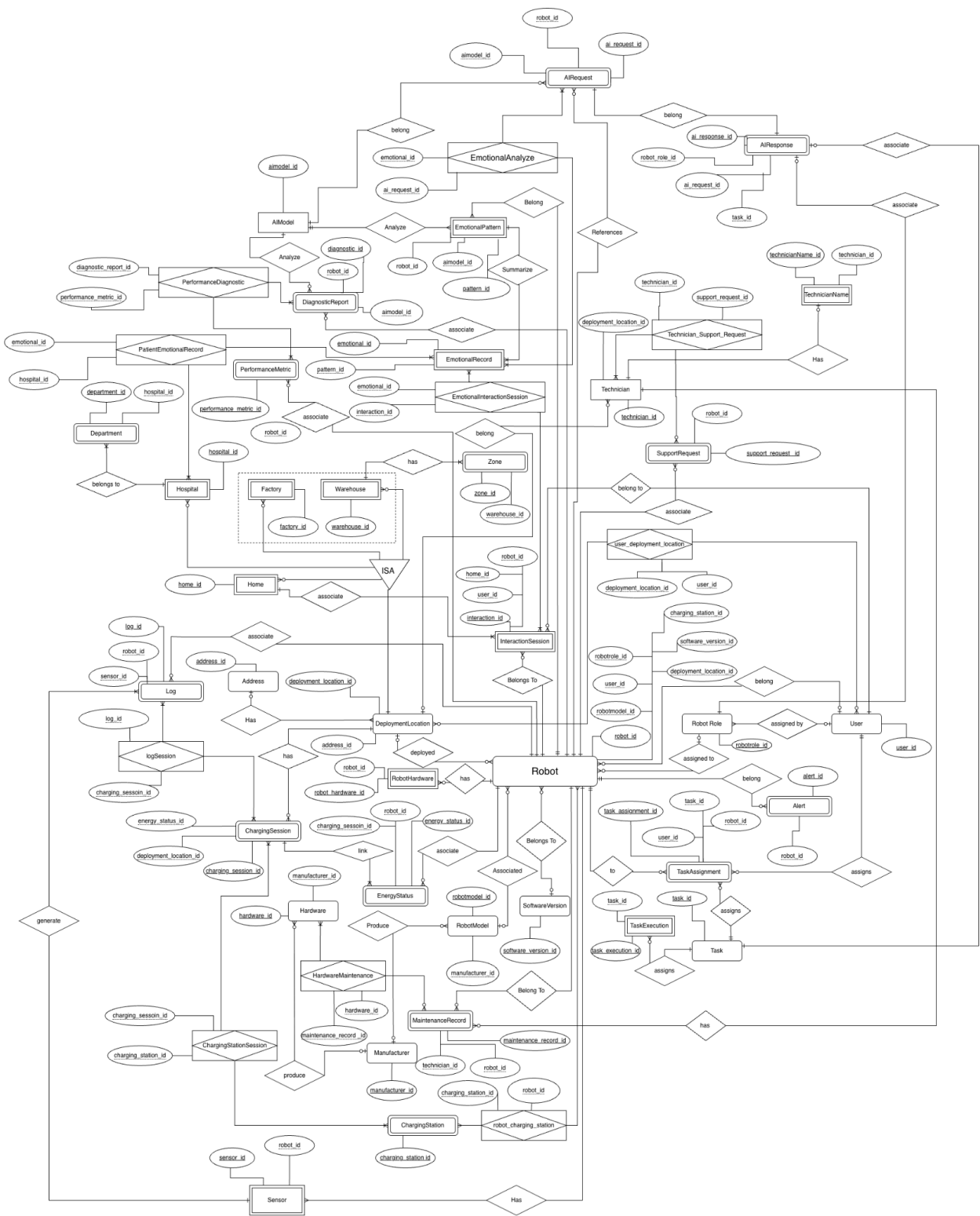
- 23.3 start_time: timestamp
- 23.4 end_time: timestamp
- 24. **SupportRequest:** Weak
 - 24.1 support_request_id: key, numeric (PK)
 - 24.2 robot_id: numeric (FK)
 - 24.3 issue_details: text
 - 24.4 time_reported: timestamp
- 25. **AIModel:** Strong
 - 25.1 aimodel_id: key, numeric (PK)
 - 25.2 model_name: alphanumeric
 - 25.3 service_type: alphanumeric
 - 25.4 version_label: alphanumeric
- 26. **AIRequest:** Weak
 - 26.1 ai_request_id: key, numeric (PK)
 - 26.2 aimodel_id: numeric (FK)
 - 26.3 robot_id: numeric (FK)
 - 26.4 requested_at: timestamp
- 27. **AIResponse:** Weak
 - 27.1 ai_response_id: key, numeric (PK)
 - 27.2 ai_request_id: numeric (FK)
 - 27.4 task_id: numeric (FK)
 - 27.5 role_id: numeric(FK)
 - 27.6 response_payload: text
 - 27.7 generated_at: timestamp
- 28. **Hospital:** Weak (Subclass)
 - 28.1 hospital_id: key, numeric (PK)
 - 28.2 deployment_location_id: numeric (FK, unique)
- 29. **Warehouse:** Weak (Subclass)
 - 29.1 warehouse_id: key, numeric (PK)
 - 29.2 deployment_location_id: numeric (FK, unique)

- 30. **Home:** Weak (Subclass)
 - 30.1 home_id: key, numeric (PK)
 - 30.2 deployment_location_id: numeric (FK, unique)
- 31. **Factory:** Weak (Subclass)
 - 31.1 factory_id: key, numeric (PK)
 - 31.2 deployment_location_id: numeric (FK, unique)
- 32. **Department:** Weak
 - 32.1 department_id: key, numeric (PK)
 - 32.2 hospital_id: numeric (FK)
 - 32.3 department_name: alphanumeric
- 33. **EmotionalPattern:** Weak
 - 33.1 emotional_pattern_id: key, numeric (PK)
 - 33.2 robot_id: numeric (FK)
 - 33.3 ai_model_id: numeric (FK)
 - 33.4 pattern_summary: text
- 34. **MaintenanceRecord:** Weak
 - 34.1 maintenance_record_id: key, numeric (PK)
 - 34.2 robot_id: numeric (FK)
 - 34.3 technician_id: numeric (FK)
 - 34.4 RecordedAt: timestamp
 - 34.5 Description: text
- 35. **Technician:** Strong
 - 35.1 technician_id: key, numeric (PK)
 - 35.2 deployment_location_id (FK)
- 36. **TechnicianName:** Weak
 - 36.1. TechnicianNameID(PK):key,numeric
 - 36.2. FirstName :Text
 - 36.3. LastName:Text
 - 36.4. TechnicianID(FK)
- 37. **User_Deployment_Location:** Associative
 - 36.1 UserDeploymentLocationID: key, numeric (PK)

- 36.1 deployment_location_id: numeric(FK)
- 36.2 user_id: numeric(FK)
- 38. **EmotionalAnalyze**: Associative
 - 37.1 EmotionalAnalyzeID: key, numeric (PK)
 - 37.2 emotional_id: numeric(FK)
 - 37.3 ai_request_id: numeric(FK)
- 39. **PerformanceDiagnostic**: Associative
 - 39.1 PerformanceDiagnosticID: key, numeric (PK)
 - 39.2 diagnostic_report_id: numeric(FK)
 - 39.3 performance_metric_id: numeric(FK)
- 40. **EmotionalInteractionSession**: Associative
 - 40.1 EmotionalInteractionSessionID: key, numeric (PK)
 - 40.2 emotional_id: numeric(FK)
 - 40.3 interaction_id: numeric(FK)
- 41. **Technician_Support_Request**: Associative
 - 41.1 TechnicianSupportRequestID: key, numeric (PK)
 - 41.2 support_request_id: numeric(FK)
 - 41.3 technician_id: numeric(FK)
- 42. **RobotChargingStation**: Associative
 - 42.1 RobotChargingStationID: key, numeric (PK)
 - 42.2 chargin_station_id: numeric(FK)
 - 42.3 robot_id: numeric(FK)
- 43. **logSession**: Associative
 - 43.1 logSessionID: key, numeric (PK)
 - 43.2 charging_session_id: numeric(FK)
 - 43.3 log_id: numeric(FK)
- 44. **HardwareMaintenance**: Associative
 - 44.1 HardwareMaintenancelD: key, numeric (PK)
 - 44.2 hardware_id (FK)
 - 44.3 maintainence_record_id (FK)

- 45. **ChargingStationSession**: Associative
 - 45.1 ChargingStationSessionID: key, numeric (PK)
 - 45.2 charging_session_id: numeric(FK)
 - 45.3 charging_station_id: numeric(FK)
 - 46. **PatientEmotionalRecord**: Associative
 - 46.1 PatientEmotionalRecordID: key, numeric (PK)
 - 46.2 emotional_id: numeric(FK)
 - 46.3 hospital_id: numeric(FK)
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EER Diagram(Normalize)



Database Schema Diagram

